

RECTANGULAR PERIODIC POINTS AND COMMUTATION TRANSFER SPECTRA FOR ONE-VERTEX 2-GRAPH SHIFTS

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ABSTRACT. For the two-sided path action of a finite one-vertex 2-graph, we reduce rectangular-sublattice fixed points to a finite commutation problem. At fixed horizontal period a , a single-red-letter transfer matrix $M_{\theta,a}$ satisfies

$$|\text{Fix}(\sigma^{(a,0)}) \cap \text{Fix}(\sigma^{(0,b)})| = \text{tr}(M_{\theta,a}^b).$$

Consequently the vertical zeta function inside the horizontal-period- a subsystem is $\det(I - zM_{\theta,a})^{-1}$. An explicit binary pair has the same vertex matrices and hence the same strictly-positive directional entropy, but different rectangular fixed-point data; this gives an entropy-blind obstruction to action-preserving $\mathbb{Z}^2$ conjugacy. The argument is factorization-theoretic, and an exhaustive finite control independently checks all 24 binary commutation bijections through periods $a, b \leq 3$.

1. INTRODUCTION

Higher-rank graphs package commuting factorizations into a category equipped with a degree functor. Their path spaces carry natural $\mathbb{Z}^k$ actions [3, 4]; entropy for the canonical path shifts has been related to vertex matrices and their Perron–Frobenius data [7]. In the one-vertex rank-two case, however, the vertex matrices remember only the numbers of blue and red generators. They do not remember the commutation bijection that fills each coloured square.

This note isolates a finite invariant that does remember that bijection. If u is a blue word of length a and v a red word of length b , unique factorization writes $uv = v'u'$. A rectangular periodic path exists exactly when the boundary words close, $uv = vu$. Holding a fixed and commuting one red letter at a time turns this condition into a closed walk in a finite multigraph on the m^a blue boundary words. This yields the trace formula and the determinant zeta identity.

The scope is deliberately narrower than the established periodicity and orbit-equivalence theories. Higher-word commutation criteria already play a central role in rank-two graph periodicity [2], while groupoid methods give broad orbit-equivalence frameworks [1]. Most directly, recent work identifies full Mealy bijections with one-vertex rank-two graphs and two-sided Wang configurations, and relates aperiodicity to period-free configurations and anti-tori [5]. We neither replace those results nor claim the first commutation/period dictionary. Our residual statement is quantitative: an exact rectangular fixed-point transfer formula and an explicit entropy-blind obstruction.

2. ONE-VERTEX RANK-TWO PATH SHIFTS

Let $E = \{e_0, \dots, e_{m-1}\}$ and $F = \{f_0, \dots, f_{n-1}\}$. Fix a bijection

$$\theta : E \times F \longrightarrow F \times E, \quad \theta(e_i, f_j) = (f_{j'}, e_{i'}),$$

and impose the relations $e_i f_j = f_{j'} e_{i'}$. The resulting category Λ_θ is the one-vertex 2-graph with degree $d(e_i) = (1, 0)$ and $d(f_j) = (0, 1)$. The defining property is unique factorization: whenever $d(\lambda) = p + q$, there are unique paths μ, ν with $d(\mu) = p$, $d(\nu) = q$, and $\lambda = \mu\nu$ [3].

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Let $X_\theta = \Lambda_\theta^\Delta$ be the two-sided path space, viewed as degree-preserving functors from

$$\Delta_2 = \{(p, q) \in \mathbb{Z}^2 \times \mathbb{Z}^2 : p \leq q\}$$

to Λ_θ . Translation gives a $\mathbb{Z}^2$ action $(\sigma^g x)(p, q) = x(p + g, q + g)$ [4]. For $a, b \geq 1$ set

$$\text{Per}_\theta(a, b) = \text{Fix}(\sigma^{(a,0)}) \cap \text{Fix}(\sigma^{(0,b)}), \quad P_\theta(a, b) = |\text{Per}_\theta(a, b)|.$$

Thus $P_\theta(a, b)$ counts paths fixed by the rectangular sublattice $a\mathbb{Z}(1, 0) + b\mathbb{Z}(0, 1)$; it is not the fixed-point count of the single element (a, b) .

For $u \in E^a$ and $v \in F^b$, unique factorization gives unique $v' \in F^b$ and $u' \in E^a$ such that $uv = v'u'$. Write

$$\Theta_{a,b}(u, v) = (v', u'), \quad \hat{\Theta}_{a,b} = \tau \circ \Theta_{a,b},$$

where $\tau(v', u') = (u', v')$. Thus $\hat{\Theta}_{a,b}$ is a self-map of $E^a \times F^b$.

3. COMMUTING BOUNDARY WORDS

Theorem 3.1 (rectangular boundary theorem). *For every $a, b \geq 1$, restriction to the lower and left boundary gives a bijection*

$$\text{Per}_\theta(a, b) \longleftrightarrow \{(u, v) \in E^a \times F^b : uv = vu\} = \text{Fix}(\hat{\Theta}_{a,b}).$$

Proof. Let $x \in \text{Per}_\theta(a, b)$, and put $u = x(0, (a, 0))$, $v = x(0, (0, b))$. Horizontal periodicity identifies the red path on the right boundary with v , and vertical periodicity identifies the blue path on the top boundary with u . The two factorizations of $x(0, (a, b))$ are therefore uv and vu , so $uv = vu$.

The pair (u, v) determines the fundamental rectangle. Indeed, the path uv has degree (a, b) , and unique factorization determines every subpath whose endpoints lie in that rectangle. Periodicity then determines all translates. This proves injectivity.

Conversely, suppose $uv = vu$. The words $u^r v^s$, $r, s \geq 0$, form a compatible family: commutation makes $u^r v^s$ the prescribed-degree prefix of $u^{r'} v^{s'}$ whenever $r \leq r'$ and $s \leq s'$. Unique factorization therefore produces a one-sided path that is periodic by $(a, 0)$ and $(0, b)$. Translate a finite subpath far enough into the nonnegative quadrant and use these two periods to define it; a common larger translate proves that the definition is independent of the choice. The resulting two-sided path lies in $\text{Per}_\theta(a, b)$ and has boundary (u, v) . $\square$

The theorem separates two roles. Local coloured-square compatibility is already built into Λ_θ ; the sole global obstruction on a torus is closure of its two boundary words. It also avoids quotienting by spatial translation: $P_\theta(a, b)$ counts based periodic paths, as an ordinary fixed-point count should.

4. THE COMMUTATION TRANSFER MATRIX

Fix $a \geq 1$. For $u \in E^a$ and $0 \leq j < n$, write

$$uf_j = f_{\alpha_a(u, j)} \beta_a(u, j), \quad \beta_a(u, j) \in E^a.$$

Define the $m^a \times m^a$ nonnegative integer matrix

$$(1) \quad M_{\theta, a}(u, w) = \#\{j : \alpha_a(u, j) = j \text{ and } \beta_a(u, j) = w\}.$$

The condition $\alpha_a(u, j) = j$ is essential: the red label must survive its passage across the blue boundary.

Theorem 4.1 (trace formula). *For all $a, b \geq 1$,*

$$P_\theta(a, b) = \text{tr}(M_{\theta, a}^b).$$

Proof. Write $v = f_{j_1} \cdots f_{j_b}$ and set $u_0 = u$. Commute the red letters successively:

$$u_{t-1}f_{j_t} = f_{\ell_t}u_t, \quad 1 \leq t \leq b.$$

Then unique factorization gives $uv = f_{\ell_1} \cdots f_{\ell_b}u_b$. Equality with vu holds exactly when $\ell_t = j_t$ for every t and $u_b = u$. Thus commuting boundary pairs are in bijection with length- b closed walks in the labelled multigraph whose adjacency matrix is (1). Summing the diagonal entries of $M_{\theta,a}^b$ proves the formula. $\square$

Corollary 4.2 (fixed-horizontal-period vertical zeta). *As a formal power series,*

$$Z_{\theta,a}(z) := \exp \left(\sum_{b \geq 1} \frac{P_{\theta}(a,b)}{b} z^b \right) = \frac{1}{\det(I - zM_{\theta,a})}.$$

The number of points in $\text{Fix}(\sigma^{(a,0)})$ with least positive vertical period b is

$$Q_{\theta}(a,b) = \sum_{d|b} \mu(d) P_{\theta}(a,b/d).$$

Proof. The determinant identity follows from $-\log \det(I - zM) = \sum_{b \geq 1} \text{tr}(M^b) z^b / b$. The second formula is ordinary Möbius inversion in the vertical period, with μ the number-theoretic Möbius function. Here the horizontal period is only required to divide a . $\square$

This is a one-variable zeta function inside a fixed horizontal-period subsystem. We do not call it a full $\mathbb{Z}^2$ zeta function.

5. AN ENTROPY-BLIND CONJUGACY OBSTRUCTION

An action-preserving conjugacy carries every subgroup fixed set to the corresponding fixed set. Hence the full rectangular profile $\{P_{\theta}(a,b)\}_{a,b \geq 1}$ is a $\mathbb{Z}^2$ -conjugacy invariant.

Consider $m = n = 2$. For the identity factorization

$$e_i f_j = f_j e_i,$$

one has $M_{\text{id},a} = 2I_{2^a}$ and therefore

$$(2) \quad P_{\text{id}}(a,b) = 2^{a+b}.$$

Now take the explicit four-cycle

$$(3) \quad e_i f_j = f_{i \oplus j} e_{1-i}, \quad i, j \in \{0, 1\}.$$

Equivalently,

$$\begin{aligned} e_0 f_0 &= f_0 e_1, & e_0 f_1 &= f_1 e_1, \\ e_1 f_0 &= f_1 e_0, & e_1 f_1 &= f_0 e_0. \end{aligned}$$

If $\theta(e_i, f_j) = (f_{j'}, e_{i'})$ and $\pi(i, j) = (i', j')$, then this specific rule has $\pi = (00 \ 10 \ 01 \ 11)$. The rule is fixed explicitly because the numerical conclusion below is not asserted for every four-cycle. In the word orders $(0, 1)$ and $(00, 01, 10, 11)$, respectively,

$$M_{\text{cyc},1} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad M_{\text{cyc},2} = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix}.$$

Thus

$$P_{\text{cyc}}(1, 1) = 0, \quad P_{\text{cyc}}(2, 2) = 8,$$

whereas (2) gives 4 and 16.

Both examples have the same one-vertex coordinate matrices [2] and [2]. For completeness, fix $p = (r, s)$ with $r, s \geq 1$. Restriction to the bi-infinite p -diagonal gives a conjugacy

$$(X_{\theta}, \sigma^p) \longrightarrow (\Lambda_{\theta}^p)^{\mathbb{Z}}, \quad x \longmapsto (x(kp, (k+1)p))_{k \in \mathbb{Z}}.$$

Strict positivity makes $\{kp : k \in \mathbb{Z}\}$ cofinal in both coordinate directions, so unique factorization reconstructs every finite subpath. Conversely, the one-vertex condition lets arbitrary degree- p blocks concatenate. Since $|\Lambda_\theta^p| = 2^{r+s}$, both examples have

$$h_{\text{top}}(\sigma^{(r,s)}) = (r + s) \log 2, \quad r, s \geq 1.$$

This direct argument is consistent with the broader positive-semigroup entropy theory [7]. Nevertheless their rectangular fixed profiles differ.

Corollary 5.1. *Strictly-positive directional entropy and the two vertex matrices do not classify binary one-vertex 2-graph shifts up to action-preserving $\mathbb{Z}^2$ conjugacy.*

6. FINITE AUDIT AND SCOPE

The accompanying exact control enumerates all 24 bijections $\theta : \{0, 1\}^2 \rightarrow \{0, 1\}^2$ and every $1 \leq a, b \leq 3$. For each of the 216 triples it compares:

- (1) direct block commutation of boundary words;
- (2) the trace of $M_{\theta,a}^b$; and
- (3) an independent enumeration of all 2^{2ab} blue/red edge labellings of the (a, b) torus, checking only the local square relation.

All 216 comparisons agree. The third implementation constructs neither boundary words nor transfer matrices, so it guards against a shared coding error in the first two. The calculation is regression evidence; the general proof is [theorems 3.1](#) and [4.1](#).

Several nearby theories remain outside the claim surface. We do not claim the definition of a higher-rank path action, the general positive-semigroup entropy formula, the periodicity criterion of Davidson–Yang, a realization theorem for general two-dimensional SFTs [6], or a complete conjugacy invariant. The Mealy/square-complex/anti-torus dictionary and qualitative period-free criteria of Pask [5] are also returned to their owner. Axis and mixed-sign directional entropy and general finite k -graphs are left open. Within those boundaries, the transfer matrix provides a concrete finite interface between categorical factorization and rectangular periodic dynamics.

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